MAINS MASTER PROGRAM (MMP) 2024
ENVIRONMENT-4
SOLID WASTE, PLASTIC WASTE, E-WASTE ETC.

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2. MUNICIPAL SOLID WASTE

- **Introduction**
  - » Solid waste is the **unwanted or useless solid materials generated from human activities** in residential, industrial or commercial areas.

- As per **Indiawaterportal.org** the total MSW generated in urban India has been estimated at **68.8 million tons per year** (TPY). It is expected to reach 165 million tonnes by 2030.
  - » But the Solid **waste collection efficiency in India is around 70%** at present, while it is 100% in many developed countries.

- **Problems of unscientific MSW disposal ->** Untreated, unprocessed and indiscriminately dumped waste causes air, water and soil pollution which have adverse impact on health situation.

- **Factors for increasing Solid Waste in India**
  - » **Population, Urbanization, Increasing Per-Capita Income**
  - » Increased Consumerism, Use and throw culture
  - » **Plastic waste** -> non availability of good alternative
  - » **Technology change** -> Increasing electronic waste
  - » **COVID-19** also led to shooting up of domestic hazardous waste.

- **Proper Solid waste management**
  - » It includes a number of processes including **segregation, collection and treatment and disposal** in an environmentally sound manner.
  - » The local authorities are responsible for the development of infrastructure for collection, storage, segregation, transportation, processing and disposal of MSW

- **Key Steps taken for Proper Solid Waste Management**:
  - » **Swachh Bharat Mission** is a step towards making India clean.
    - ▪ To carry forward the progress made, Swachh Bharat Mission (SBM-U) 2.0 has been launched on 1st Oct 2021 for a period of five years, upto 1st Oct 2026, with a vision of **achieving Garbage Free Status for all cities through 100% source segregation, door to door collection and scientific management** of all fractions of waste, including safe disposal in scientific landfills.
      - • It is also aimed at remediation of all legacy dumpsites and converting them into green zones.
  - » **Solid Waste Management Rules, 2016**

- **Key Concerns and Problems of Solid Waste Management System in India**
  - » Conflicting data on Quantum of waste generated
  - » **Segregation hasn't happened:** Further, ULBs have failed to establish systems and technologies required for segregation, collection, and processing of different categories of waste.
    - ▪ **Waste to Energy** system also face problems due to lack of segregation.
  - » **Complete lack of focus on Domestic Hazardous waste** (sanitary pads, syringes, medicines, paint boxes, batteries, mercury containing substances etc.)
    - ▪ Not exhaustively covered in SWM Rules 2016
    - ▪ Lack of infrastructure for proper processing

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- Lack of awareness among people about the damages these may cause

» **Open dump sites** outside the city are the most common way of dumping wastes.

» **Local authorities** face resource crunch both in terms of funds and human resource.

» **Most Indian villages don't have any waste management infrastructure**: A study released by non-profit Pratham Education Foundation in Sep 2022.

» **R&D and indigenization** of waste processing technologies is missing.

- **Way Forward related to Solid Waste**

» **Better implementation of Solid Waste Management Rules, 2016**

▫ **Set Targets:** Draw up a phase-wise timetable for achieving the source segregation.

▫ **More resources to ULBs**: Promote door to door collection of waste by subsidizing waste collection by ULBs.

» **Better Monitoring: Proper digitalization in waste collection and disposal operations** -> this will ensure real time monitoring of collection and transportation operations.

» **Enhancing waste treatment capacities:** We need to go for more R&D in **Waste to Compost** and **Waste to Energy** technologies.

» **Special Focus on Domestic Hazardous Wastes**

▫ Exhaustively cover them under the SWM Rules 2016.

▫ Bring Domestic Hazardous waste under EPR (currently only plastic waste and E-Waste are covered under EPR in India)

» **Promote the 4R Principle** through behavioural change:

▫ Waste reduction strategy can be incorporated by each of us whether at home or at work by following the 4 Rs principle [Reduce, Reuse, Recycle, Recovery or Reclaim].

- **Reduce**: Every citizen can begin with lesser packaging, more durable & refillable items; carry your own shopping bags; avoid disposable items; and reduce the use of plastics. In office, paper work can be cut down and more e-mail communication could be used.
- **Reuse**: Donate old books, clothes, phones etc.
- **Recycle**: Segregate waste for better disposal and purchase recycled/ green products.
- **Recovery of Reclaim:** Better implementation of biomining

▫ Community awareness and a change in people's attitude towards solid waste and their disposal can go a long way in improving India's SWM system.

3. VARIOUS WAYS OF DEALING WITH SOLID WASTE – ADVANTAGES AND LIMITATIONS

- **Introduction** (Class discussion)

1. **Open Dumping, Landfills and Sanitary Landfills**

» **Advantage**: Waste limited to well defined area; Reduces contact between waste and environment.

» **Disadvantages** - Open dumps get exposed to natural elements, stray animals and birds and may cause air pollution, water pollution and soil pollution.

2. **Thermal Treatment**

» **Incineration plants (Waste to Energy Method)**

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- Incineration is combustion of waste in the presence of oxygen. Waste gets converted in CO_2, Water Vapor and Ash along with **heat**.
- **Advantages** - reduction in volume; kills many diseases causing germs.
- **Limitations -** Air pollution -> Health issues; Climate Change

» **Pyrolysis**
- Here material is exposed to **very high temperatures** in an **inert (oxygen less) environment.** The material decomposes due to the limited thermal stability of chemical bonds of material, which disintegrates.
- Pyrolysis is thus a **thermo-chemical** treatment, which can be applied to any organic (carbon-based) product. It produces volatile products and leaves a solid residue enriched in carbon, char.

» **Plasma Arc Gasification (PAG)** process
- It is a waste treatment technology that uses a **combination of electricity and high temperature** to turn municipal waste (garbage or trash) into **usable by-products without combustion**.
  - It converts the organic waste into gas that contains all its chemical and heat energy and converts the inorganic waste into an inert vitrified glass called **slag**.
  - This process reduces the volume of waste reaching the landfills and also generate electricity.

3. **Biological Treatment Methods** - Use of microorganisms

» **Bio-Gasification**: Biological decomposition of organic matter of biological origin under un-aerobic condition is done to produce **methane** and other secondary gases.

» **Composting**: The o**rganic waste is converted into compost through decomposition**.
- Compost is rich in nutrients and can be used as soil conditioner, a fertilizer, addition of vital humus and humic acids and as a natural pesticide in soil.
- It can also be used for erosion control, land and sea reclamation, wetland construction, and as landfill cover.

» **Vermiculture/Vermicomposting**: It is the process of making compost through decomposition process. But here, decomposition is done by using various species of worms, usually red wigglers, white worms, and other earthworms.

» **Bioremediation**: It involves use of bio-culture or microorganisms to degrade organic waste and contaminants that pose environmental and human risks. Here the environment is altered to stimulate the growth of micro-organisms and degrade pollutants. The organic waste is eventually converted into soil.
- **Various approaches** - Biostimulation; Bioaugmentation; a combination of both etc.

4. **Biomining**

» Biomining involves use of separator machines or large sieves to separate waste material of different sizes, thereby obtaining soil, plastic, wood and metal components in isolation for appropriate processing.

- **Conclusion**: The most appropriate way of dealing with waste depends on various factors including type of waste, advancement of technology etc. But with a combination of the above given method, it is possible to dispose of solid waste in an environment friendly manner.

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4. ISSUE OF LANDFILLS AND LANFILL FIRES

- **Introduction**
  - A landfill fire occurs when waste disposed of in a landfill ignites and spreads. In landfills that don't cover their waste with daily cover, biological decomposition creates substantial heat and can cause material in the landfills to spontaneous combust.
  - Landfill fire has become a regular phenomenon now with the issue regularly been in news. Deonar landfill fire in Feb 2016 got India's focus and global pressure on this issue.

- **Causes -** Methane Gas; Sabotage; collection of scrap metals; difficult to extinguish.

- **Impact**
  - **Ground Water and River Water** Pollution: Leachate from these landfills is not only contaminating ground water but are also reaching Yamuna River.
  - **Other concerns due to landfills** -> Air Pollution (methane); Odour Pollution; Wastage of Resources; breeding ground for diseases.
  - **Prolonged exposure** to compounds such as **dioxins** which are carcinogenic.

- **Way Forward**
  - **Short term measures**
    - Increasing security at the landfill sites
    - CCTV cameras including night vision cameras (were installed at Deonar landfill sites in Mumbai)
    - Setting up waste to energy plants
    - Preparing fire safety plans
    - Stationing fire tenders at the site
    - Spreading awareness among waste collectors
  - **Long Term Measures**
    - **Limiting the waste** going to dumping grounds
      - Segregation of wastes
      - Reusing the reusable waste
        - Most construction waste can actually be reused
        - Biodegradable wastes can be converted into composts or biogas
      - Following the guidelines for Municipal wastes, solid wastes and e-wastes Rules
    - Limiting the production of waste
      - Massive education program for general public
      - Training programs for rag pickers
    - Solid waste treatment plant mandatory at all dumping sites
    - Scientific processing of solid waste
    - **Recycling is the key – Biomining**

5. SOLID WASTE MANAGEMENT RULES, 2016

- In 2016, The Environment ministry had revised the solid waste management rules after 16 years
- Salient Features
  - » **Extended Beyond Municipal Areas** - Covers urban agglomerations, census towns, notified townships, areas under control of railways, airports, airbases, port and harbour, SEZ etc.
  - » **Source Segregation** of waste has been mandated to channelize the waste to wealth by recovery, reuse, and recycle.
    - ▪ Waste generators have to segregate the waste in **three streams** - Wet (biodegradable; Dry (plastic, paper, glass, metal etc.) and Domestic Hazard wastes (diapers, napkins, empty containers etc.)
  - » **Street vendors** to keep **separate containers** for separate wastes
  - » **Sanitary napkins and diapers** manufacturers or brand owners explore the possibility of using **recyclable material** in the product and shall provide a **pouch or wrapper** for disposal of each napkin or diapers along with packet of their sanitary products.
    - ▪ Educate masses in wrapping and disposal of their products.
  - » The rules emphasized on **integration of waste pickers/ rag pickers and waste dealers** in the formal system by state governments, SHGs or any other group to be formed.
  - » **Ban on open throwing burning or burying**; **Provisions for User Fee** for waste collectors and 'Spot Fine' for Littering and non-segregation.
  - » **Provisions for Bulk and institutional generators** -> directly responsible for segregation and sorting the waste and manage in partnership with local bodies;
  - » **All manufacturers of disposable products such as tin, glass, plastics packaging etc. or brand owners who introduce such products** in the market should provide **necessary financial assistance to local authorities** for the establishment of waste management system.
  - » **The Biodegradable waste** -> processed through composting, bio-methanation etc.
  - » **Promoting Waste to Energy**
    - ▫ Industrial units within 100 km of Solid waste RDF plants should get at least 5% of their fuel from them.
    - ▫ Non-recyclable waste with high calorific value (1500 K/cal/kg or more) should not be disposed of and should only be utilized for refuse-derived fuel or by giving away the feedstock for preparing refused derived fuel.
    - ▫ **High calorific wastes** shall be used for co-processing in cement or thermal power plants.

6. WASTE TO ENERGY PLANTS (W2E PLANTS)

- A waste to energy plant is a waste management facility **that combusts wastes to produce electricity**.
- **Two ways of using Municipal solid waste** in Waste to Energy Plants
  - i. Which burns all municipal wastes
  - ii. Which burns RDF (refuse derived waste): RDF is produced from combustible components of MSW removing out components such as glass, etc. The waste is shredded, dried and baled and then burned to produce electricity.
- Advantages of W2E Plants
  1. **Clean method of waste disposal:** It drastically reduces the quantity of waste
  2. **Energy Resource**
  3. **Reduces pollution of water and air:** Unnecessary landfill fires don't occur
  4. **Financial Benefits**

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- **Profitability**: W2E can be a profitable business if the right technology is employed with optimal processes. Government incentives can multiply the benefits.
- **Government incentives**: GoI already provides a lot of incentives for W2E plants in the form of feed in tariffs and capital subsidies.

- **Main Constraints faced by India's waste to Energy Plants**
  1. **Still a new concept,** so not a lot of R&D has gone in the sector
  2. **Import dependency** of the technology
  3. **Western Technology** not suitable for Indian scenario
  4. **The cost of project** (especially based on bio methanation technology) are high as critical equipment for a project is required to be imported.
  5. **Segregated waste is generally not available** at plant site
  6. **Lack of financial resources** with **municipal corporations/urban local bodies**
  7. **Lack of conducive policy guidelines from state governments** in respect of allotment of land, supply of garbage and power purchase/ evacuation facilities
  8. **Pollution** from plants cause concern for employment and health for individual in particular
- **How MNRE is helping develop the W2E plants**
  1. **Financial Assistance**
     - **Interest subsidies** for commercial projects
     - Assistance on the capital cost for demonstration projects that are innovative
     - Financial assistance for municipal corporation for supplying garbage free of cost at the project site and for providing land.
  2. **Incentives given to state nodal agencies** for promotion, co-ordination and monitoring of such projects
  3. **R&D Promotion:** Financial assistance for carrying out studies on W2E projects, covering full cost of such studies
  4. **Assistance given in terms of training courses, workshops and seminars and awareness generation**
- **Criticisms of W2E**
  » **High levels of pollution**: A WtE unit actually converts the waste into invisible particulate and gaseous pollutants in the atmosphere leading to heightened inhalation hazard.
  » **The domestic and Municipal waste** is also amongst the least efficient fuel
  » Though proponents quote rosy pictures of a few W2E plants still operating in Japan, Sweden and Amsterdam; they simply ignore the fact that thousands and perhaps millions of W2E plants around the world have been closed in the past three-four decades.
- **What can be done to promote W2E more?**
  » **Improved Collection** (segregated waste etc)
  » **Promote PPP** in W2E plants
  » Amend **electricity Act 2003** - to provide for mandatory buying of electricity from these plants by discoms.
  » **More R&D** in new technology for Indian scenario
  » More funds to be provided for urban local bodes
  » **Focus on Alternatives** as W2E may not be sustainable solution
     - Avoid creation of large landfills; let waste treatment happen within the Municipal area; Spread awareness about proper segregation mechanism;

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7. PLASTIC POLLUTION

- **Introduction**
  - » Plastic is a lightweight, hygienic and resistant material which can be molded in wide range of applications and is cheaply manufactured. Because of these reasons, since the 1950s, the production of plastic has outpaced almost all other materials.

- **Plastic overshoot Day Report**: By **Earth Action (EA) (July 2023)**
  - » On July 28, 2023, the Earth saw its first Plastic Overshoot Day: The point at which the amount of plastics exceed the global waste management capacity - As per Swiss based research consultancy **Earth Action** (EA).
    - **Nearly** 68 million tonnes of additional plastic waste will end up in nature in 2023.
    - **India** is among the 12 countries of the world including China, Brazil, Indonesia, Thailand, Russia, Mexico, USA, Saudi Arabia, the DRC, Iran and Kazakhstan, which are responsible for 52% of the world's mismanaged plastics.
    - Under current scenario, despite pledges and increased waste management capacity, increased production of plastics will lead to global plastic pollution tripling by 2040.

- **India** produces around **10 million tonnes of plastic** per year of which **around 5 million tonnes** is rendered waste every year. Therefore, its crucial that this waste is properly managed.

- **Harmful Impacts of Plastic Pollution**
  - » **Physical Pollution:** Pieces of plastics interact with living bodies and ecosystems. It causes physical harm through ingestion, choking and entanglement hazards to wildlife on land and in ocean.
  - » **Chemical Pollution:** A number of chemicals used in plastic are toxic and carcinogenic and are responsible for infertility, recurrent miscarriage, feminization of male foetuses etc.
  - » **Environmental Impacts:**
    - ▫ **Plastisphere**: Sometimes called the 'Plastisphere', **bacteria, viruses and other life colonize the surface of plastic waste**, creating distinct communities and population structure.
    - ▫ They may also contribute in growth of **invasive species**. For e.g., more than 80% of invasive species in the Mediterranean may have arrived on floating plastic waste.
  - » **Health and Social Impact:** Health losses, welfare losses -> unusable parks, Sewage Blocking -> Malaria, Dengue etc.
    - ▪ As per a study published by World Wildlife Foundation (WWF) in 2019, an average human may be ingesting as much as **5 grams** of plastic every week. This is because 1/3rd of the plastic waste that is getting generated ends up in nature, especially water, which is largest source of plastic ingestion.
  - » **Economic Impact**
    - ▪ **Visual pollution** negatively impacts the tourism sector.
    - ▪ Further, **future cost** of removing these plastics from nature is higher than the cost of preventing the littering today.
  - » **Exacerbate disasters like floods** - an important cause of urban floods.
  - » **Exacerbates Climate Change:** Plastics are 80% carbon and more than 99% of plastics use crude oil, fossil gas or coal as feedstock.

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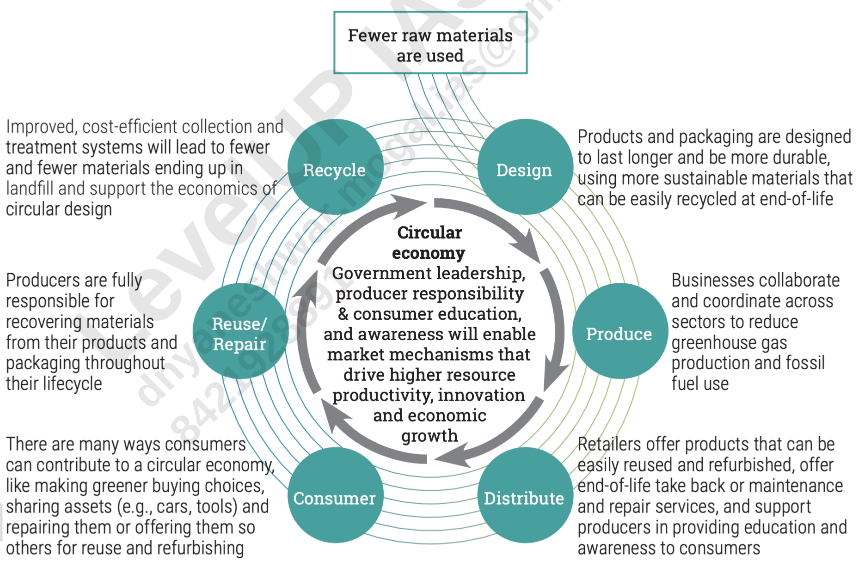
- **Some Recent Efforts towards reducing Plastic Waste and ensuring better plastic waste management**
    a. **Plastic Waste Management Rules 2016** (as amended in 2021)
    b. **Guidelines on EPR for Plastic Packaging** under 'Plastic Waste Management Rules, 2016: Notified in Feb 2022
    c. **Strengthening of waste management infrastructure** through Swatch Bharat Mission.
    d. **Promotion of Alternatives**:
        - CPCB has already provided certificates to more than 300 manufacturers of **compostable plastics**.
        - India Plastic Challenge - Hackathon conducted for development of innovative alternatives to SUP.
    e. **Steps for Effective Monitoring** (E.g. an SUP Public Grievance App, an SUP compliance Monitoring Portal etc.)
    h. **Awareness Generation**:
        - **Mascot 'Prakriti'** has been launched to spready awareness about how small lifestyle changes can play a big role in environmental sustainability.
    i. **Promoting Alternative uses of Plastic Waste**: For e.g.
        - waste plastic is being used as replacement of coking coal (by upto 1%) in steel manufacturing.
        - MoRT&H have also issued guidelines for use of plastic waste in road construction.
    j. **WWF-India and CII** have joined hands to develop a platform to promote a circular system for plastics. The new platform is called, the '**India Plastic Pact**'.

- **International Efforts**
    a. **Steps towards Plastic Pollution Treaty**: In 2022, the UN member states agreed to start negotiating new global treaty to end plastic pollution. By July 2024, 4 rounds of negotiation had taken place:
    b. **Awareness and Education:** The theme of **World Environment Day, 2018** was "**Beat Plastic Pollution**" and it focused on increasing awareness related to plastic pollution across the world.
    c. **Other International Initiatives which deal with Plastic Waste**
        i. **#Clean Seas campaign** of UN Environment
            - To reduce and eliminate the use of single use plastic, cosmetics and micro-plastic sources
        ii. **Stockholm Convention on POPs**
            - It is an international environment treaty that came into force in 2004 and aims to restrict or eliminate the production of PoPs.
        iii. **Honolulu Strategy by UNEP** aimed at reduce **marine debris**
            - The Honolulu Strategy is a framework for a comprehensive and global collaborative effort to reduce the ecological, human health, and economic impacts of marine debris worldwide. This framework is organized by a set of goals and strategies applicable all over the world, regardless of specific conditions or challenge.
- **Key Challenges faced by Plastic Waste Management in India**
    - Problems like **lack of people's participation, source segregation, absence of a suitable alternatives** etc are increasing the load on plastic management sector.

- **<u>Domination of Informal sector</u>:** Around 70% of the plastic waste management industry is <u>informal in nature</u> and yet no major effort towards formalizing the industry has been pushed in recent times.
- **Plastic Waste management** has been put under a **5% GST** bracket <u>hurting the formal sector</u>, which already lacks a concrete action plan.
- **Implementation of Plastic waste management rules** have been poor in all aspects.
    - In the past too, <u>many states in India have banned plastic at state level</u>. However, these bans have a <u>very little impact on ground</u>.
    - A CAG Report in Dec 2022 highlighted that <u>MoEF&CC has no mechanism to assess the collection and safe disposal</u> of plastic waste. This has led to <u>poor implementation of the 2016 rules</u> as well as the <u>EPR guidelines</u>.
- Since waste management is the responsibility of the **<u>local bodies</u>**, the **<u>resource constraints</u>** with our local bodies in terms of **human resource and infrastructure** is another major challenge.
- **COVID-19 pandemic** has increased the plastic pollution burden drastically - masks, gloves, sanitizer bottles, PPE kits etc.

- **Way Forward for India**
    - **Improving the Waste Management System**

    - **Promoting Circular Economy**: As per a recent report by UNEP titled - "*Turning Off the Tap: How the world can end plastic pollution and create a circular economy*" - <u>Global Plastic Pollution can reduce by 80% by 2040 if countries and companies make deep policy and market shifts using existing technologies and shift to **circular economy**</u>.



    - **Reducing the Usage of Plastic**
        - **Focus on Packaging Innovation -> Promote Eco-friendly alternatives:** This will reduce the <u>use of plastic</u> for packaging purpose which has expanded in recent years, especially due to e-commerce.
            - Companies need to <u>invest more in R&D</u> to find sustainable alternatives.

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- **Taxing Plastic Production** can also increase the cost of plastic and thus help in reducing the usage
- **Awareness generation** among end users.

- Steps to **implement EPR** -> innovative solutions such as offset mechanism and deposit refund schemes. The producers/generators may also form **waste cooperatives** and employ informal waste pickers if offset mechanism is in place.
- **Strengthening Local Bodies -> More resources with local bodies** can be ensured by support from producers. This resource can contribute in more sustainable waste management system by ensuring better infrastructure.
- **Social Awareness and Public Pressure**
- **International collaboration and coordination** -> A Global pact like Montreal Protocol and Paris Agreement for the reduction of plastic production and usage.

- **Conclusion**
    - With a worldwide crisis due to plastic waste, India has to find a way to curb its plastic pollution at the earliest and that is only possible when all the stakeholders take the responsibility of ensuring minimization, reuse and recycling of plastic to the maximum.

1) PLASTIC WASTE MANAGEMENT RULES, 2016 (AS AMENDED IN 2021): KEY PROVISIONS

- **Min thickness** of plastic carry bags has been increased to **120 microns from 31st Dec 2022** (after the 2021 amendment to the rules)
- **Ban on Several Single Use Plastic:** The manufacture, import, stocking, distribution, sale and use of several single use plastics (e.g ear buds with plastic sticks, plastic sticks for balloons, plastic plates, cups, glasses etc.) including polystyrene and expanded polystyrene commodities shall be prohibited wef from 1st July 2022. (as per the 2021 amendment)
    - **Plastic Packaging Waste**, which is not covered under the phase out of identified single use plastic items, shall be collected and managed in an environmentally sustainable way through the **EPR** of producer, importer and Brand Owner (PIBO), as per the Plastic Waste Management Rules, 2016.
- **Expand the coverage to rural areas**. The earlier regulations only covered urban municipal areas.
- **Introduces Extended Producer Responsibility** for producers and generators of Plastic Waste
- **Shopkeepers and Vendors** can only use plastic carry bags which have been properly labelled and marked for use or else there will be imposition of fines.
- **ULB and Panchayats** have been provided with the responsibility of establishing and operating waste management systems.
- The **Land Department** (or any department with business allocation of land allotment with state governments) should allocate land for establishing waste management facilities.
- **Gainful usage of Plastic waste** has also been promoted in road construction, waste to oil conversion etc.

2) GUIDELINES ON EPR FOR PLASTIC PACKAGING UNDER PLASTIC WASTE MANAGEMENT RULES, 2016: NOTIFIED IN FEB 2022

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- **Mandatory Registration** on the centralized portal developed by CPCB of entities such as producer, importer, brand owners and waste processors of plastic.
- The guidelines categorizes SUPs in **4 categories and provides targets for minimum level of recycling**.
- **Environmental Compensation** shall be levied based upon polluter pay principle, with respect to non-fulfilment of EPR targets.
- **Sale and Purchase of surplus EPR certificates are allowed** -> this has thus set up **market mechanisms** for plastic waste management.
- **Focus on Digitization**: Implementation of EPR will be done through a customized online platform which would act as the digital backbone of the system. It will allow tracking and monitoring of EPR obligations and will reduce the compliance burden.
- Further, to ensure monitoring on fulfilment of EPR obligations, the guidelines have prescribed **a system of verification and audit of enterprises.**
- Producers, importers, & brand owners, may operate **schemes such as deposit refund system** or **buy back** or **any other model**.
- CPCB shall **constitute a committee under chairpersonship of Chairman, CPCB** that shall be responsible for recommending measures to MoEF&CC for the effective implementation of EPR that shall included amendments to the EPR guidelines.
- **Promote development of new alternatives** to plastics

- **Significance:**
  - Promote Development of Alternatives
  - Gives boost to formalization and further development of plastic waste management sector.
  - By operationalizing EPR, the amendment implements the **'Polluter Pay Principle". It** aims to hold manufacturers and producers of ecologically unsustainable plastic items financially and socially accountable for the pollution these materials cause.
  - Online Monitoring mechanism -> Increase accountability
  - Market based approach -> Promotes Ease of Doing Business.
  - The amendment demonstrates India's political will to address the challenge of plastic pollution.

- **Concerns/Challenges**
  - The biggest challenge will be to implement highly ambitious recycling and reuse targets in a country where various existing waste management rules haven't been implemented.

8. GLOBAL PLASTIC TREATY NEGOTIATIONS

- **Why in news?**
  - 2nd Session of Intergovernmental negotiation Committee (INC) on plastic pollution was held in Paris in June 2023.

- **Background:** In 2022, the UN member states agreed to start negotiating new global treaty to end plastic pollution. Now its is crucial that the treaty that is finalized is ambitious and effective enough to truly address the plastic crisis.
  - The Intergovernmental Negotiation Committee (INC) on Plastic Pollution is in the process of developing "an international legally binding instrument on plastic pollution, including in the marine environment"
  - As of July 2024, 4 negotiation meetings, for the new treaty has taken place (INC-4 took place in April 2024 in Ottawa).

- **Why is a global Treaty on Plastic Pollution required?**
  i. **Plastic Pollution is a global problem** which requires global solution. Most of the plastic is being dumped into oceans. This is eventually converting into micro-plastics, entering food chain and affecting everyone.
  ii. Plastic pollution is harmful to wildlife and biodiversity which is impacting everyone.
  iii. **Increased International Cooperation** will be feasible through a global treaty.
  iv. The treaty may **set global target for reduction**
  v. A global treaty may make the fight against plastic pollution more fair -> by giving higher responsibility to developed economies and giving more time to under developed countries.

- **Way Forward:**
  » **Fast Track the** Negotiation to finalize the treaty quickly (unlike the agreements under UNFCCC which took many years of negotiations.
  » The treat should have **plans to reduce production and consumption o**f plastics and chemicals used in plastics, especially by businesses.
  » It should have provisions to **regulate priority sectors like packagin**g which use unsustainable amounts of plastics.
  » Along with reduction, **mechanism to ensure transparency** with respect to production, consumption and import/export of plastic and plastic waste has to be created and nurtured at a global level.

- **Conclusion1:**
  » A strong global plastic treaty would be a major step forward in the fight against plastic pollution. It will not only help environment, but will also be crucial for human health and our future.

- **Conclusion2:**
  » Agreeing to a global agreement on plastic pollution by the end of 2024 would mark one of the most significant environment al decisions and would be first of its kind agreement to unite the world around a shared goal to end plastic pollution.

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9. E-WASTE

- **Introduction**
    - » E-waste is a popular name for electrical and electronic equipment (EEE) discarded after their end of life'.

- **Annual waste Output**
    - » Global E-waste monitor (published by UN University) estimates that **53.6 million tonnes (7.3 kg per capita)** of e-waste was generated world over in 2019.
        - ▪ This is an increase of 21% in just five years.
        - ▪ This is expected to go to **74.7 Mt** by 2030.
    - » India is the fifth largest generators (around 2 mt annually) of E-waste after China, USA, Japan and Germany.

- **E-waste generation in India is expected to grow rapidly in the coming future** (income, urbanization, changing technology, import (legal or illegal), poor quality equipment, power surge issues etc.)

- **Harmful effects of e-waste:** -
    - » Hazardous and toxic heavy metals - mercury, cadmium etc;
    - » Ozone Depleting Substances.
    - » High Global Warming Potential gases;
    - » Unscientific extraction - Air Pollution, Water Pollution and Soil Pollution;
    - » Severe negative health impacts - hampers central and peripheral nervous system, bran development, kidney, reproductive system etc.

- **Laws / Rules**
    - a. **E-Waste Management Rules, 2022** notified by MoEF&CC on 2nd Nov 2022. It replaced the E-Waste (Management) Rules, 2016 and became effective from 1st April 2023. These rules will have new EPR regime for e-waste recycling.

- **Key Concerns regarding e-waste management in India**
    - a. **Poor e-waste management infrastructure** in India: Present recycling and collection facilities in India are completely insufficient for the quantum of waste that is getting generated.
    - b. **Lack of skilled labor to handle e-waste recycling:** For e.g. nearly 50,000 people make living out of Seelampur's e-waste but almost all of the mare from the unorganized sector and unskilled..
    - c. **Poor working conditions in unorganized e-waste management sector ->** Extraction uses crude mechanisms polluting air, water and soil, Involvement of children.
    - **d. Formal vs informal recycler conundrum**
        - ▪ **Difficulty of formal recyclers: Extra Regulations; Expenditure; Poor Reach**
        - ▪ **Difficulty of informal recyclers -> lack infrastructure, skilled workforce, pollute environment and endanger health of workers.**
        - ▪ **Solution - Integrate**
    - e. **Ineffective enforcement of 2022 rules:** EPR work is hardly visible and states have not done much to ensure safety, health and skill of the workers.
    - f. **Absence of Awareness** about e-waste management among common masses is adding to increased generation of the waste.

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   g. Weak regulatory framework to control both **exports and imports** of e-waste and ensuring their environmental and sound management.

- **Way forward**
   a. **Inventorization** of e-waste generated annually; Proper planning and disposal can be effective only after having proper data.
   b. **Implement EPR provisions strictly:** Both **manufacturers and vendors** should be made financially, physically and legally responsible for their products.
   c. **Quarterly collection drives** through both rural and urban areas should be conducted by government and local bodies.
   d. **Awareness generation** is the key in e-waste management:
      - Consumers as well as other stakeholders, including producers, waste pickers and e-waste recyclers have to understand the problems and correct management techniques
      - **Reduce, Reuse and Recycle**
         - **Minimize the generation of E-waste** - Ensure longevity of the products, stress on the less toxic and easily recoverable alternatives. **E-Fasting** can also minimize e-waste.
         - **Reuse and Recycling** are also crucial here. Recovery of metals, plastic, glass, and other materials reduces the magnitude of e-waste.
   e. **Producer Responsibility organizations** needs to network with unorganized waste-pickers.
   f. **Promoting R&D Effort** at safe metal recovery from circuit boards etc.
      - Currently, this is mostly done by large companies outside India.
   g. There is a need to **integrate unorganized and organized sectors** to benefit from the best of both worlds
      - Unorganized sector -> better reach
      - Organized sector -> better infrastructure, better regulation.
   h. **Clear Regulatory instruments** to control both **exports and imports** of e-waste and ensure their environmentally sound management.
   i. **E-Waste** is an '**urban mine**' as it contains several precious, critical, and other non-critical metals that, if recycled, can be used as secondary materials.
      - The **value** of raw materials in the global e-waste generated in 2019 is equal to approximately **$57 billion USD**.

- **In conclusion** it can be said that e-waste is becoming a ticking time bomb that can go off disastrously at any moment. Safe disposal practices, consumer awareness, cooperation between organized and unorganized sector and strict enforcement of laws and rules are needed to solve the problem. Capacity building and increasing public awareness are the need of the time.

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1) E-WASTE MANAGEMENT RULES, 2022

a. **E-Waste Management Rules, 2022** notified by MoEF&CC on 2nd Nov 2022. It will replace E-Waste (Management) Rules, 2016 and will be effective from 1st April 2023. These rules will have new EPR regime for e-waste recycling.

- **Key Features:**
  - The rules are applicable to every manufacturer, producer, refurbisher, dismantler and recycler. They are all required to **register on portal** developed by CPCB.
  - **Expansion of Schedule 1:** Now **106 Electrical and Electronic Equipment (EEE)** have been included under EPR regime.
  - **Solar Power Panels** - included in the rules.
  - **Fixed well defined target:**
    - Producers of the notified EEE have been given annual e-waste recycling targets based on the generation from the previously sold EEE or based on sales of EEE as the case may be.
    - Target may be made stable for 2 years and starting from 60% for the year 2023-2024 and 2024-25; 70% for the year 2025-26 and 2026-27 and 80% for the year 2027-28 and 2028-29 and onwards.
  - **EPR Certificates**: Provision for generation and transaction of EPR certificates to provide a market-based approach to reach targets.
  - **Reduction of hazardous substances** in manufacturing of EEE: It mandates that EEE shall not contain lead, mercury and other hazardous substances beyond the maximum prescribed concentration.

A) MANAGEMENT OF SOLAR PV MODULES/CELLS HAS BEEN ADDED IN CHAPTER V OF THE SAID RULES.

- As per these rules, every manufacturer and producer of solar photo-voltaic modules or panels or cells shall:
  i. Ensure registration on the portal;
  ii. store solar photo-voltaic modules or panels or cells waste generated up to the year 2034-2035 as per the guidelines laid down by the Central Pollution Control Board in this regard;
  iii. file annual returns in the laid down form on the portal on or before the end of the year to which the return relates up to year 2034-2035;
  iv. ensure that the processing of the waste other than solar photo-voltaic modules or panels or cells shall be done as per the applicable rules or guidelines for the time being in force;
  v. ensure that the inventory of solar photo-voltaic modules or panels or cells shall be put in place distinctly on portal; and
  vi. comply with standard operating procedure and guidelines laid down by the Central Pollution Control Board in this regard.

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10. SOLAR WASTE

- **Introduction**:
  - » **Solar waste** refers to waste generated during the manufacturing of solar modules and waste from the field (project lifetime)
  - » **E-Waste Management Rules,** 2022 includes solar waste in the definition of E-waste

- **Current Situation of India**: Report: "**Enabling a circular Economy in India's Solar Industry - Assessing the Solar Waste Quantum**
  - » The analysis was done by MNRE and Council on Energy, Environment and Water (CEEW), a climate think tank in March 2024
  - » **Generation of solar waste**: 100 Kilotons in FY22-23. It is expected to reach 600 kt by 2030 (this report is referring to end of life waste)
    - ▪ The **current solar capacity** of India was 66.7 GW as of March 2023 which is expected to go to 293 GW by 2030.
    - ▪ Therefore, management of solar waste has to be given very high priority.
  - » **5 States -** Rajasthan, Gujarat, Karnataka, TN, and Andhra Pradesh - will be responsible for around 67% of the waste produced.
  - » **Critical Minerals**: Discarded modules also contain critical minerals such as Silicon, Copper, tellurium, and Cadmium.

- **Key Recommendations of the Report**:
  - » Maintain a **comprehensive database** of the installed solar capacity, which would help in estimating solar waste in the following years.
  - » **Promote Recycling**: Incentivize recyclers, and push stakeholders to effectively manage the growing solar waste.
    - ▪ **Focus on creating market for solar recyclers**
  - » **Shift towards high-value recycling**:
    - ▪ Conventional recycling involves mechanical process like crushing , sieving, and shearing of the waste. This method is able to recycle glass, aluminium, and copper, more valuable materials like silver and silicon can't be recovered through this method.
    - ▪ High value recycling involves mechanical, thermal and chemical processes, to recycle the module. It is also able to recycle silver and silicon.
- **Other way forward**:
  - » **R&D**: Innovation in design may have an impact on the type of waste they generate; technology advancements will be significant in reducing the impact of renewable energy waste

- **Conclusion:**
  - » With India's ambitious target of 280 GW by 2030, it is time to formulate a sound solar waste management policy.

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11. BATTERY WASTE MANAGEMENT RULES, 2022

- MoEF&CC, Government of India published the Battery Waste Management Rules, 2022 on 24th August, 2022 to ensure environmentally sound management of waste batteries. It replaced the 2001 rules.
- The rules cover **all types of batteries**, viz. Electric Vehicle batteries, portable batteries, automotive batteries and industrial batteries.
- The rules function based on the concept of **Extended Producer Responsibility (EPR)** where the producers (including importers) of batteries are responsible for collection and recycling/refurbishment of waste batteries and use of recovered materials from wastes into new batteries
    - EPR mandates that all waste batteries to be collected and sent for recycling/refurbishment, and its prohibits disposal in landfills and incineration.
    - The rules will enable setting up a mechanism and **centralized online portal** for exchange of **EPR certificates** between producers and recyclers/refurbisher to fulfil the obligations of producers.
- The rules promote **setting up of new industries and entrepreneurship in collection and recycling/refurbishment** of waste batteries.
- **Mandating the minimum percentage of recovery of materials from waste batteries** under the rules will bring new technologies and investment in recycling and refurbishment industry and create new business opportunities.
- Prescribing the **use of certain amount of recycled materials in making of new batteries** will reduce the dependency on new raw materials and save natural resources.
- On the principle of Polluter Pays Principle, environmental compensation will be imposed for non-fulfilment of Extended Producer Responsibility targets, responsibilities and obligations set out in the rules. The funds collected under environmental compensation shall be utilised in collection and refurbishing or recycling of uncollected and non-recycled waste batteries.

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12. CONSTRUCTION AND DEMOLITION WASTE

- **Introduction**
  - » With development comes massive generation of construction & demolition (C&D) waste and a number of concerns associated with them.

- **Key Concerns**
  - » **No proper estimate** on the volume of C&D waste being generated in the country.
    - ▪ **CPCB -> 25-30** million tonnes; CSE Analysis - 530 MT in 2013
  - » **Illegal dumping on vacant sites** (under fly overs; besides lakes & rivers; low lying areas etc.)
  - » **Lack of Awareness** regarding dumping rules, recycling advantages etc. People are still apprehensive about the quality of recycled product.
  - » **Air Pollution:** Adds to high amount of **PM** content in air.
  - » **Water logging** due to debris further adds to adverse impact on public health and environment.
  - » **Wetlands are threatened** -> illegal dumping leads to ponds, lakes being encroached upon.
  - » **Mixing of C&D with Municipal waste** led to
    - ▪ Reduction in the effectiveness of composting and bio-methanation.
    - ▪ Reduces the calorific value and combustibility of the MSW
    - ▪ Presence of MSW with C&D waste reduces the quality of recycled C&D waste.
  - » **Recycling Industry** is in very nascent stage in India. (According to CSE - only 1% recycling)
  - » **Large exploitation of natural resources**

- In **2016,** government for the first time came up with **Construction and Demolition Waste Management Rules**, 2016. These rules are aimed at promoting recovering, recycling and reuse of the waste generated through C&D.
  - » **Mandatory segregation** of C&D waste into four types - concrete, soil, steel and wood, plastics, bricks and mortars.
  - » Deposit it at **collection centers** or hand it over to **processing facilities**.
  - » All stakeholders responsible for waste **disposal** (be it small scale generators, the municipal body or the government)
  - » It makes debris **recycling mandatory**
  - » **Illegalizes the dumping** of waste outside designated areas.
  - » **Waste processing** authorities -> should have authorization from SPCB and should be located far away from habitation.
  - » For **effective monitoring** of the rules, specific roles have been allocated to **CPCB**, the **Bureau of Indian Standards** (BIS), the **Indian Road Congress** (IRC) and Central Ministries.
  - » **Land Department** - Provide land for storage processing and recycling of C&D waste
- **Way Forward**
  - » **Strict implementation of the rules** -> Presently even government bodies are violating the rules
  - » **Reuse and Recycle** of C&D waste needs to expand.
  - » **Promote Recycle Industry** which promotes innovation and new products and their uses.
  - » **Learn from International System**
    - ▪ **Singapore** was recycling 98% of the construction waste by 2007.
    - ▪ Most **EU members** have set goals for recycling C&D waste that range from 50% to 90% of their C&D waste production.
    - ▪ **California** state have also promulgated an ordinance which calls for 50% recycling of C&D waste.

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13. BIOMEDICAL WASTE

- **Introduction:** Biomedical waste is any waste that is generated during the diagnosis, treatment or immunization of human being or animals or in research activities pertaining thereto or in the production or testing of biologicals.

- **Sources of Biomedical waste -** Health Care Facilities (HCFs), Research institutes, Blood banks, mortuaries, biotech institutes, industrial production units, butcheries etc.

- **Quantity of Waste Produced in India:** According to ASSOCHAM India in 2022 GENERATED 775 TPD OF BMW.
- To grapple this increasing stream of bio-medical waste, as of Feb 2019, there **are 200 Common Biomedical Waste Treatment Facilities** (CBWTFs) in India. This is definitely inadequate for health facilities in 750 districts of the country.

- **Problems that can be created by Bio-Medical Waste**
  - **Environmental Pollution** - all air, water and soil
  - **Unpleasant smell**
  - **Transmission of diseases** - e.g. COVID-19.
  - **Injuries** due to sharp edges of the waste
  - Disposables and drugs being **repacked and sold** by unscrupulous elements.

- **Biomedical Waste Management Rules, 2016 (new rules released in March 2016** and amended in 2018 and 2019**)**
  - **Pretreatment** of laboratory waste, microbiological waste, blood samples and blood bags through disinfection or sterilization **on site** should be carried out as prescribed by WHO or NACO (National Aids Control Organization).
  - **Waste classification in four categories instead of 10** to improve the segregation of waste sources
  - **Phased discontinuation of chlorinated plastic bags**, gloves and blood bags
  - **Bar-code system to classify disposal of bags of containers having BMW**
  - **More Stringent standards** have been prescribed for incinerators to reduce the pollution
  - **States to provide land** for setting up common biomedical waste treatment and disposal facility.
- **Key Issues Associated with Biomedical Waste Management in India**
  - **Poor implementation** and **monitoring** of BMW Rules 2016 and its 2018 amendment.
  - **Too Much to Handle: Poor infrastructure:**
    - As per a study by Down to Earth, 22 out of 35 states and Uts generate more biomedical waste than they can handle.
    - For e.g. The UT of Lakshadweep doesn't have a single common biomedical waste treatment facility.
  - **Under-reporting of waste generated** and handling capacity
  - Lack of **Awareness** among various sections of staff about BMW handling and rules to be followed.
  - **Lack of segregation practices**, result in mixing of hospital wastes with general waste
  - **Lack of funds** with CBWTFs -> hinders capacity aggregation and improvement.
- **Way forward**
  - **Inventorization** of total number of HCFs and the quantity of waste generated.
  - **Stringent legal** action against the violators of the rules (HCF or the CBMWTFs)

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- **Training and Awareness Programs** for healthcare personnel needs to be conducted for proper segregation and better waste management.
    - BMW Management rules should also be made part of medical education curriculum.
- **Encourage self-regulatory mechanisms** for monitoring and implementation for waste management
- **Well timed sufficient resources (funds and human resource)** should be ensured for CBMWTFs to improve their capacity and infrastructure.

- **Conclusion**
    - Combating the problems of BMW has brought forward a number of humanitarian and environmental challenges and therefore, needs immediate responsiveness for our common world.

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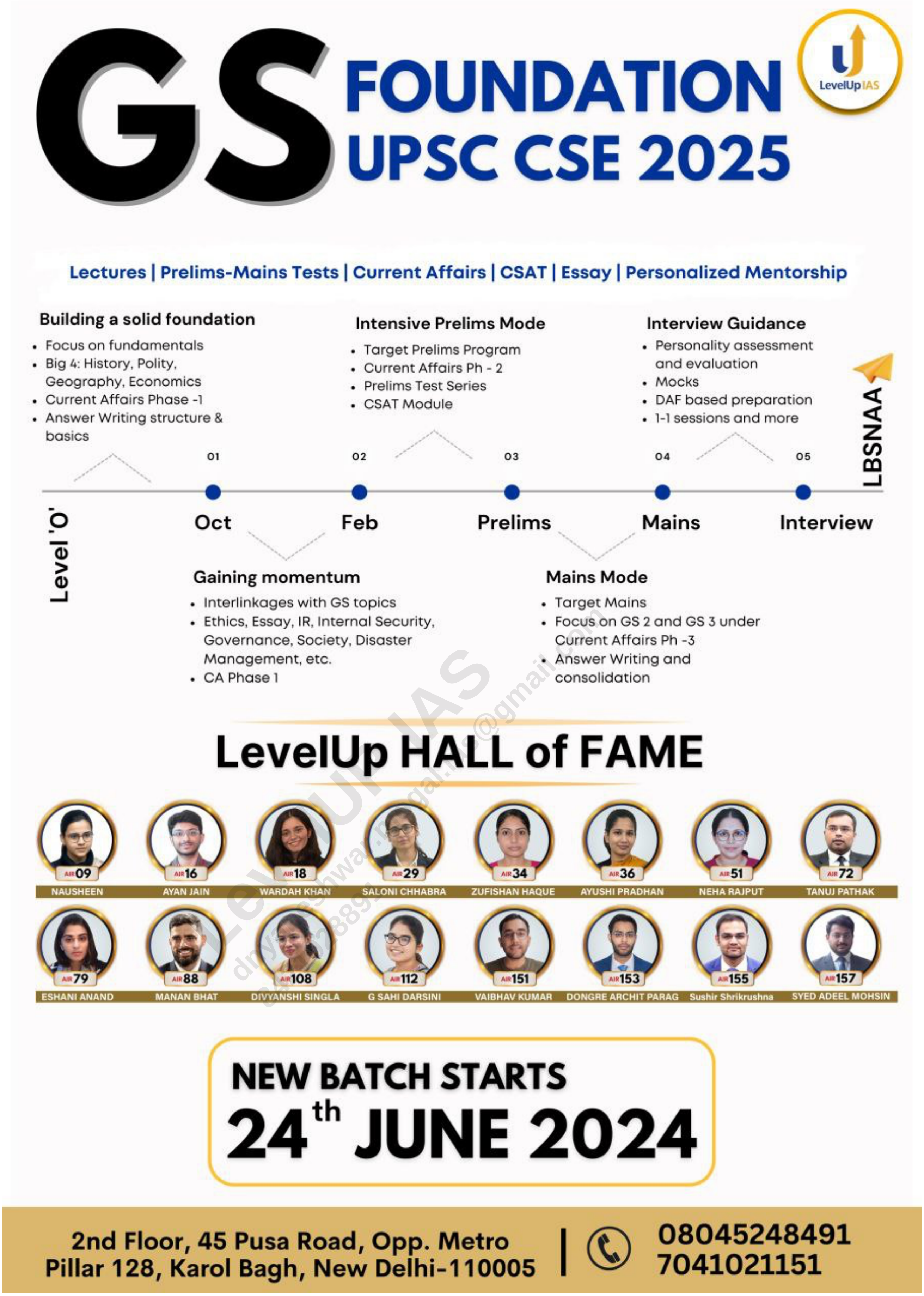
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